The Psychosocial Factors That Affect The Smoking Habits Of Filipino Adolescents

A Research Proposal

Dayta, Dominic B.

2013-45527

Background

Reports by the Global Adult Tobaccco Survey (GATS; WHO, 2010) reveal that 28.3 percent or over 17.3 million of the country's population aged 15 years and above are classified smokers in 2009. The habit seems to exist predominantly amongst men: 47.7 percent of the male population are smokers (consuming on average 11.3 cigarettes per day), compared to only 9 percent of the female population (consuming 7 on average only 7). The combined percentage for the general population might not look so big, but it implies that now almost a third of the population have gotten into the habit of smoking, with an average initiation age of just 17 years for men and 19 years for women. Despite its known adverse consequences and efforts by the national government to curb the sales and use of cigarettes, the country's tobacco situation seems nowhere close to improving.

The present administration has resorted to drastic measures such as an all-out public smoking ban in the hopes of changing the picture. Yet, despite hopeful projections by the government that the ban, if executed properly, could lower smoking prevalence in the country by as much as 4 percent (Kapoor & Dela Cruz, 2016), such a form of prohibition isn't likely to produce favourable results, as it simply pressures current smokers away from the habit without

regards to how the habit was initiated in the first place, and the various external and internal factors that may have caused it.

For example, the World Health Organization (WHO, 2007) cites widespread marketing and low prices of tobacco products as one of the main explanatory factors for their worldwide popularity. It was observed in the same report that the number of smokers are observed to be the highest in countries such as Egypt and the Philippines, where the price of a cigarette per pack also happens to be the lowest. Similarly, influences among peers and family member have been identified to contribute in initiating an individual into smoking.

Various other social, psychological and economic factors have been shown to be associated with initiation into the cigarette habit. Moon et al (2010), Chaiton et al (2009) and Escobedo et al (1998) have shown repeatedly and with varying methodologies that depression among patients is a significant predictive factor for nonsmokers to pick up the habit. Precisely why this association exists will have to be left for in-depth psychological inquiry, but for the moment there is strong foundation at least in statistical literature to hypothesize that tobacco use can be expected to be more prevalent among subjects under strain from depression.

Bans put in place do little to nothing in curbing demand. In Owyand and Vermann (2012), it is shown that despite the spike of smoke-free ban laws in both the local and national levels in the United States, this has done very little in encouraging smokers to quit, or nonsmokers to avoid picking up a cigarette. Specifically, the study found no significant difference in the number of nonsmokers initiated into the act with or without a ban (the study refers specifically to bans enacted in municipality, workplace, and public areas). Similarly, the number of smokers that continue the habit is not statistically significant from zero with or without a ban. For both scenarios, however, an enforced nationwide ban was found to be

statistically significant from zero with a negative correlation (ie, less continuation and initiation with a ban than without), but the researchers observed that the differences remained minimal for the ban to be conclusively considered effective. Owyand and Vermann ultimately conclude that there is little effect observable of a ban on the smoking behaviors of the general public.

While a comprehensive all-out ban on public smoking might indeed help curb the numbers somewhat, it can only be expected to act as a stop-gap solution to a far more complex sociological problem, as it only takes into account forcing those who are already smokers away from the habit by limiting the places in which they may practice it, but it does not take into account why they started smoking in the first place. Should the national government wish to pursue efforts that will have actual impact on tobacco statistics, these efforts should target the root factors that may be contributing to the popularity of cigarettes in the country.

The present study seeks to confirm the relationship between smoking prevalence among adolescents in the country and various social, psychological and economical indicators exhibited by individuals living in a barangay in the Philippines. Specifically, the study will delve into the association present between depression, economic standing, and peer/family influences on the smoking habits of these individuals, using a survey on a selected sample of adolescent smokers.

Related Literature

Psychosocial Factors. The relationship between psychosocial factors contributing into the onset of cigarette smoking among individuals have been explored in various previous literature. In Fagan et al (2005), using a sample of 603 American young adolescents who were then followed up with until reaching 27 years of age, found that parental employment (whether the parent classified as a blue-collar or white-collar worker), educational achievement, and

parental smoking habits were significant factors affecting the onset of cigarette smoking. Specifically, smoking was observed not only to become more prevalent but also to onset the individual much earlier among adolescents with blue-collar employed parents with low educational achievement and who were also smokers and often practiced the habit in the presence of their children.

For this study, the researchers will focus entirely on individual smokers and the various factors that affect their cigarette habits. Why this is so is due to an assertion by Flay et al (1998) that studying smoking behaviors and its association with various explanatory variables must focus on the adolescent individual (aged 10 to 19). An individual falling within the stated age range is rich with various interpersonal interactions (with peers, family, etc.) as well as intrapersonal developments that may contribute to a smoking behavior. In an article by the Paediatr Child Health titled “Age Limits and Adolescence,” characterizes the adolescent individual as undergoing a steady process of identity formation and acceptance, beginning with their initiation into much of the world outside the home), and are thus usually unstable in terms of psychology, open to outside influences, and curious to try new things.

Therefore, more than others, adolescents are more impressionable and susceptible to taking up the habit of smoking. In fact, it has been stipulated that most smokers started out in their adolescence (or perhaps even younger), and the habit simply carried on to their latter ages. In Chassin et al (1996) and Moolchan et al (2000), adolescents already initiated into cigarette smoking will likely continue the habit to very late in life. This is based on longitudinal studies done on American adolescents.

The parents' own behavior will likely affect the child's perception of smoking. To reiterate the findings of Fagan et al (2005), and in addition assertion by Caliskan (2009) that

employment level (blue- or white-collar), employment type (self-employed or hired), educational attainment, and smoking behaviors are significant factors that eventually contribute to the child picking up the habit of smoking and continuing it to the late stages of life. Gritz et al (1998) determined that children are more inclined to smoke when their parents are also – and very visibly – smoking.

Peer groups form another sociological factor that affects a child's smoking habits. Yazici and Sahin (2005) put special emphasis on the smoking habits of friends, schoolmates, and distant acquaintances in initiating an initially nonsmoking individual into the habit. Ceylan et al (2005) even goes so far as to declare it one of the most important factors why an individual would even consider picking up their first cigarette.

Smoking and Economy. Every year, 5 million people around the world die of causes directly linked with cigarette smoking – that means one person adds to the death toll every eight seconds (Caliskan, 2009). Yang et al (2009) estimates that the total death toll due to smoking will have reached 8 million globally by 2030, but pints to what may appear as an ironic fact: that 80 percent of these deaths are accounted for in the less developed countries of the world. Despite cigarette smoking being considered an expensive habit to pick up and maintain, it appears that it has become more popular in the poorer parts of the world. The focus of the study by Yang deals with Chinese urban laborers and it is hypothesized that the excess smoking prevalences may be explained by unstable working conditions and the erratic pace of city life in industral China.

Some form of vicious association seems to exist between smoking and poverty. For some reason, the neediest people in the world also happen to be the most active smokers. Thus, the researchers include economic factors in assessing the contributing variables to an adolescent's

initiation into smoking. Smoking rate appears to increase faster in countries with low and medium incomes. Those who are lower in the social and economic ladder are "more inclined to smoking." Thus, the level of education and household income are expected to associate negatively with smoking rate (Caliskan, 2009).

Depression, Economic Standing, and Smoking. It was also found that the solitude amongst Chinese laborers may have contributed to the excess smoking prevalences. That they are usually absent from home and separated from loved ones induces anxiety and despair among laborers, making the more susceptible to smoke, especially since it was found that smoking prevalence is higher among migrant workers, or those who have traveled from their home in the provinces or from different countries looking for better opportunities in Urban China (Yang et al, 2009).

It has been stipulated in various statistical literature that an association exists between depression and smoking. Moon et al (2010), Chaiton et al (2009) and Escobedo et al (1998) have shown repeatedly and with varying methodologies that depression among patients is a significant predictive factor for nonsmokers to pick up the habit. Moon, Mo and Basham (2010) showed that nonsmokers who exhibited high levels of depressive symptoms had fifty percent higher chance of becoming a smoker the following year than those with lower symptoms of depression. The study was conducted on a sample of 2,735 boys and 2,890 girls in America, all in their adolescence, with mean age of 16.04. Escobedo et al (1998) showed this same result, using longtudinal data from a sample of 7,885 adolescents aged twelve to eighteen interviewed in 1989 and 1993. After reviewing longitudinal studies on the association between smoking and depression in adolescents, Chaiton et al (2009) concludes that the data suggests a two-way association between the two.

In Hong Kong, Lam et al (2005; cited in Moon, et al, 2010) used anonymous self-reported surveys on 1,894 seventh grade students in two waves separated by a 12-month interval. Lam et al (2010) showed that students who at the first wave never smoked by showed high symptoms of depression had a 50% likelihood of becoming or having become a smoker by the second wave. Results also showed that higher levels of depressive symptoms can be observed from a participant classified as a former or current smoker.

Depression, however, would be elusive to detect, especially among respondents coming from the Philippines. Data from the World Health Organization (WHO, 2007) shows that there are only 0.41 psychiatrists in the Philippines per 100,000 of the general population. In a country where simply being in connection with a licensed psychiatrist is seen as a status symbol, using medical records in gathering a sample of respondents with depression will only produce a misrepresented picture of the general population.

Other methods of assessing an individual's mental health will have to be incorporated into the survey, by producing an index that will rate a person's psychological strain based on questions that refer to key indicators of depression. This questionnaire method will expectedly produce a tradeoff in reliability than if a professional were to assess the subject's metal health, but such a tradeoff will be favorable for allowing more respondents from different economic sectors to join the study instead of only those who have the means to consult with a doctor.

Statement of the Problem

The study aims to explore whether various psychosocial and economic variables may be used to explain smoking behavior among adolescents using multiple linear regression. Can a model be fitted on the smoking behavior of Filipino adolescents, for the purpose of prediction

given information on an individual's specific economic and psychosocial conditions?

Definition of Variables

The present study aims to confirm the presence of psychosocial and economic factors that affect the smoking habits of Filipino adolescents residing in Barangay 22 of Caloocan City, Metro Manila.

Factors to be included in the study are:

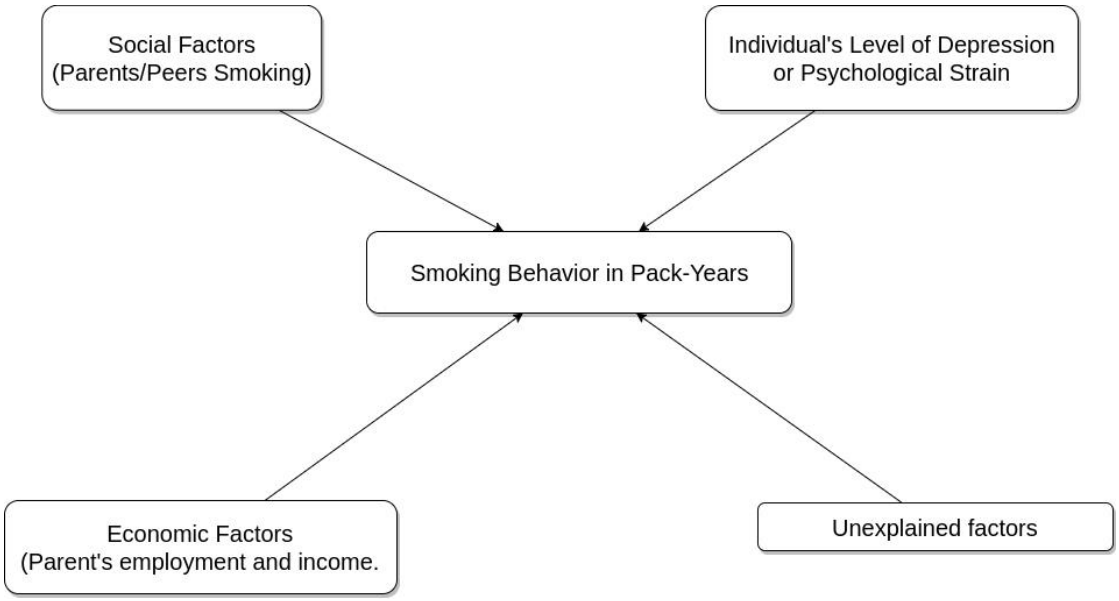
Variable Name	**Values**	**Description**
EMPLOY	0 – unemployed 1 – employed	Parents' employment status
INCOME	$[0, +\infty)$	Parents' (combined) annual income
FAMSMOKE	0,1,2,…	Number of household members that smoke
PEERSMOKE	0,1,2…	Number of peers that smoke
EDUC	0 – currently out of school 1 = currently studying	Respondent's schooling status
DEPINDEX	$[0, +\infty)$	A depression index based on a series of questions corresponding to key indicators.
ANXINDEX	$[0, +\infty)$	An anxiety index based on a series of questions corresponding to key indicators.

Computation of the DEPINDEX and ANXINDEX will be discussed in subsequent portions of the proposal. Meanwhile, the dependent variable will be the individual's smoking behavior in pack-years as described in Janjigian et al (2010) as:

PACKYEAR = [(CURRENT AGE) – (AGE STARTED SMOKING)] X [(DAILY NUMBER OF CIGARETTES SMOKED)/20]

A pack-year measures the amount that a person has smoked over time since starting out as a smoker, used in clinical studies to determine a smoker's intensity of habit as well as their risks to various diseases associated with cigarette smoking, especially lung cancer.

Schematic Diagram



The above schematic diagram has been adapted from Fagan et al (2006).

Three main classification of factors are included in the model, and these are hypothesized to have an effect on the smoking behavior of individuals. At the same time, other unknown factors or known factors not accounted for in the model (see *Limitations* below) may also be adding to the individual's smoking behavior. Note that the diagram assumes independence (non-branching) between the Social Factors, Psychological Strain, and Economic Factors, as this is the

ideal setting for a multiple linear regression model, as a problem of multicollinearity will come up otherwise. This is only assumed, however, and does not necessarily apply to the data once it has been collected. Thus, it will be checked through diagnostic checking of the regression model.

Limitations

The study will have the limitation that the respondents' true mental condition cannot be calibrated with absolute reliability, although the use of adapted questionnaires will help maximize the reliability of the findings. It remains paramount that a person's suffering from mental strain, depression, anxiety, and isolation be determined by licensed medical professionals.

However, considering the limitations of the researchers in not affording the assistance of such a licensed medical professional to screen each respondent entered into the study, the questionnaires will have to suffice. On the other hand, the researchers cannot simply make use of existing records since psychiatric help is not as widespread in the country as it is in more developed areas of the world, and in the inclusion of economic factors, it remains the interest of the researchers to draw in respondents from various economic sectors, not only those who have the means to consult with doctors.

At least as far as the topic being proposed in its current form, one commonly discussed facor contributing to smoking behavior and demand will not be included in the study: the exposure of individuals to marketing of cigarettes. This is because a majority of the population can now be expected to have encountered in at least one form an advertisement of cigarettes, and it is only a matter of quantifying the level of exposure one has to cigarette marketing. As no literature or methodology has been found or proposed that will allow the researchers to quantify the level of exposure to cigarette marketing for each individual, this will not be included among

the factors considered in the study. Similarly, cigarette prices and availability has shared much of the blame for widespread smoking across the globe, but as the study will consider only one sample of individuals in one time-frame (cross-section), there is no point in including the mentioned variables into the study although there is strong backing from statistical literature that they will contribute significantly to smoking behavior.

Statistical Analysis

Sample Size Determination. Nonprobability sampling in the form of quota sampling will be used for the present study. From results by Knofczynski and Mundfrom (2008), a total size of 100 is adequate for use in multiple linear regression problems. Houses will be selected at random from a list procured from the Barangay, and systematic selection will be done until a hundred individual adolescent smokers have been surveyed.

Multiple Linear Regression. As the end-goal of the study is to produce a model that will help predict the smoking behavior of an individual based on their specific psychosocial and economic conditions, the data will be subjected to a regression analysis making use of classic linear regression on mixed continuous and categorical variables. Specifically, the hypothesized model:

$$\text{PACKYEARS}_i = \beta_0 + \beta_1\ \text{EMPLOY}_i + \beta_2\ \text{INCOME}_i + \beta_3\ \text{FAMSMOKE}_i + \beta_4\ \text{PEERSMOKE}_i + \beta_5\ \text{EDUC}_i + \beta_6\ \text{DEPINDEX}_i + \beta_7\ \text{ANXINDEX}_i + \varepsilon_i$$

Other descriptive methods will be utilized in the statistical analysis, such as comparing the mean pack-years observed from the respondents between employment levels, educational status, the

smoking behavior of parents and peers; meanwhile, a correlation analysis will be done pairwise to find any relationship between the smoking behavior of individuals in pack years and their depression index scores and their family incomes.

Below are some dummy tables for outputs to be expected in the analysis:

Table 1: Coefficient estimates for the full model (Overall p-value: XXXX; R-square: XXXX)

Coefficient	***Estimate***	***Standard error***	***p-value***
Constant	XXX	XXX	XXX
EMPLOY	XXX	XXX	XXX
INCOME	XXX	XXX	XXX
PRNTSMOKE	XXX	XXX	XXX
PEERSMOKE	XXX	XXX	XXX
EDUC	XXX	XXX	XXX
DEPINDEX	XXX	XXX	XXX
ANXINDEX	XXX	XXX	XXX

The p-values for the individual t-tests on the variables included in the full model will be used in removing insignificant explanatory variables, and shaping a final model containing only significant predictors of pack-years smoking behavior, whose estimates will then be displayed in the table below.

Table 2: Coefficient estimates for the final model (Overall p-value: XXXX; R-square: XXXX)

Coefficient	***Estimate***	***Standard error***	***p-value***
Constant	XXX	XXX	XXX
EMPLOY	XXX	XXX	XXX
INCOME	XXX	XXX	XXX
PRNTSMOKE	XXX	XXX	XXX
DEPINDEX	XXX	XXX	XXX
ANXINDEX	XXX	XXX	XXX

During the modeling procedure, interim estimated models will be run through a series of diagnostics that will check for the underlying assumptions for multiple linear regression. Such assumptions include: normality and homoscedasticity of error terms, as well as independence of regressors. Normality will be tested using the Jarque-Bera goodness-of-fit test, while the Breusch-Pagan test will be used to test for homoscedasticity. As for the presence of multicollinearity among the regressors, the Variance Inflation Factors will be inspected. In the process of performing model diagnostics, various charts will also be produced, such as a quantile plot of the residuals to confirm the Jarque-Bera test for normality of error terms, and residual plots to confirm the results of the Breusch-Pagan test for homoscedasticity. A table for of correlations will also help in the purpose of checking for correlations between regressors:

Table 3: Correlations of Depression and Anxiety Indices with Pack-Years and Income

	Pack-Years	***Income***	***Depression***	***Anxiety***
Pack-Years	-	XXX	XXX	XXX
Income	-	-	XXX	XXX
Depression	-	-	-	XXX
Anxiety	-	-	-	-

On the DEPINDEX AND ANXINDEX Scores. The respondents will be made to answer a questionnaire consisting of questions that refer to key indicators of depression and anxiety amongst individuals. Each item in the depression/anxiety index questionnaire can be answerable on a likert scale of five levels (to allow for neutrality), coded on a scale of 1 to 5, and then to produce an index score, the researcher will make use of principal components based on the questions referring to indicators of depression. The anxiety index is computed likewise.

Data Collection Tool

Since much of the data required in the study is straightforward, the only issue with designing the questionnaire rests on the items that refer to key indicators of depression. There might be some necessary approaches required for breaching potentially sensitive questions such as whether one's parents are unemployed/employed, or whether the individual being interviewed is currently going to school or has stopped (either temporarily or permanently). Also, in certain cases a separate interview may be necessary to have with the parents pertaining to questions the

adolescent respondent might not have an idea about, such as the family income.

On the other hand, for the questions regarding the respondent's psychological condition, the following questions have been adapted from the Goldberg Depression Questionnaire. The responses in the present version (as prototype) have been scaled to five level Likert responses. Each of these questions refer to various key indicators of depression among individuals. While depression is not a quantifiable psychological state, the Depression Index to which the responses in this questionnaire will be pooled only provides a likelihood score that the individual indeed suffers from depression. The precise sensitivity and specificity measures of the test will have to be left for inquiry and left out of the present study.

Note that the first question can be used as a screening question to identify the smokers.

Also note that the questionnaire as it is has been presented below is not the final version and is merely all the questions to be included pooled together, to provide a general picture of how the final approved questionnaire will appear.

Part I

Question (check the box corresponding with the response)	**Yes**	**No**
I am a smoker		
Parents currently employed?		
(If unemployed) Parents currently involved in business (self-employed, partnerships, etc.) earning income?		
I am actively attending school.		
At least one of my parents are smokers.		

At least one member (other than parents) of the family are smokers.		
Most of my peers are smokers.		
Question (write answer on the right)	**Answer**	
What age did you start smoking?		
How many cigarettes do you usually smoke in a day?		
Complete birthday?		
Family income? (may be asked of the parents if they are around)		
Number of household members living together?		

Part II. Respondents are asked to rate their level of agreement to the following statements with 5 signifying very strong agreement, 1 very strong disagreement, and 3 as neutral.

	5	4	3	2	1
I feel lacking in energy.					
I am generally uninterested in activities.					
I have little confidence in myself.					
I feel hopeless about many things in life.					
I usually have difficulty concentrating in activities.					
I see no point in going out and meeting with people.					
I have been losing my appetite.					

I have been waking up very late recently.					
I feel physically slowed down.					
I feel mentally slowed down.					
I feel down when I wake up in the morning.					

Part III. Respondents are asked to rate their level of agreement to the following statements with 5 signifying very strong agreement, 1 very strong disagreement, and 3 as neutral.

	5	4	3	2	1
I feel unsettled or at edge.					
I have been worrying a lot.					
I have been very irritable.					
I have been finding difficulty relaxing myself.					
I have been sleeping poorly.					
I have been having headaches.					
I have been experiencing at least one of the following: trembling, tingling, dizzy spells, sweating, diarrhea, or needing to urinate more often than usual.					
I have been worrying about my health.					
I have been worrying about the wellbeing of people I know.					
I have been having difficulty falling asleep.					

Budget

Considering the straightforward nature of the study, there are very little budget considerations. Listed below:

Expenditure	**Expected Cost**
Instrumentation	P400.00
• Papers	• P 300.00
• Staples	• P100.00
Fund for scouting and mapping	P5,000.00
Fund for interviewers' transportation	P5,000.00
Funds for respondent incentives	P5,000.00
Food Allocation	P10,000.00
Total	**P25,400.00**

Schedule of Activities

Considering a seven-week (around two months) schedule, the following may be a possible arrangement of duties per committees in performing the data collection.

Activity	**Week**						
	1	**2**	**3**	**4**	**5**	**6**	**7**
Scouting and Mapping of Locale	X						
Preparation of Instrumentation	X						
Dry Run and Mock Interviews	X						
Interviews		X	X	X	X		
Statistical Analysis + Paper Writing						X	X
Finalization of Output							X

References

(Refer to original draft.)

Additional:

Knofcynski, Gregory T. and Mundfrom, Daniel (2008). "Sample Sizes when Using Multiple Linear Regression for Prediction." *Educational and Psychological Measurement*, 2008 68:431. 11